Sagar Agarwal

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Professional Summary

B.Tech Computer Science student at BIT Mesra with experience shipping full-stack applications, AI/ML systems.

Education

Birla Institute of Technology, Mesra — B.Tech, Computer Science 2023 – 2027 | CGPA: 8.56

Technical Skills

- **Languages:** Python, Java, C, SQL, JavaScript, Dart
- **AI / ML:** TensorFlow, Scikit-learn, NLP, TF-IDF, K-Means Clustering, Neural Networks, RAG, Llama 3.1
- **Web & Mobile:** React.js, Flutter, Firebase, Google Cloud Platform (GCP), REST APIs
- **CS Fundamentals:** Data Structures, Algorithms, Object-Oriented Programming, DBMS, Operating Systems
- **Tools:** Git, GitHub, Chart.js, Twilio, Vercel

Internship Experience

Daanik Eduventures — Software Developer Intern May 2026 – Jul. 2026

- Built features for **Wealth management platform**, a fintech platform for retail investors, including [robo-advisory scenario analysis with LLM-based RAG / KYC verification via API Setu / live video consultations with Agora SDK], shipping 5+ features across 2 production deployments.

Projects

SaanchaarSetu – Interoperability Middleware [GitHub] May 2026

FastAPI · Kafka · Sentence Transformers · pgvector · Redis · Debezium · React

- Engineered a **bidirectional sync middleware** connecting **Karnataka's Single Window System** to **40+ legacy department systems** (Factories, Labour, Fire Safety, BBMP) via a **3-tier ingestion pipeline** (webhook · polling · CDC via Debezium) – zero modifications to any source system.
- Designed a **semantic schema mapping engine** using **Sentence Transformers + pgvector** with **PII scrubbing** before all AI inference, **Redis idempotency gating**, exponential-backoff retry with **Dead Letter Queue**, and a **hash-chained append-only audit log** – reducing cross-department data propagation from days to **under 5 seconds**.

GraphSentinel (AI-Powered Fraud Detection) [GitHub] July 2026

PyTorch Geometric · Neo4j · Kafka · Flink · Docker · Kubernetes

- Developed a real-time banking fraud detection system using Temporal GNNs (EvolveGCN) and LSTM Autoencoders to detect multi-hop money laundering patterns natively at scale.
- Secured **0.89 F1-score** for fraud classification and a **94% reduction in FP** compared to standard threshold-based AML rules.
- Ranked **Top 40 of 1,600+ teams** at IDEA Hackathon 2.0, currently extending the system with federated learning.

Digital Twin Market Engine — *AI Product Launch Simulator* [GitHub] Feb. 2026

React.js · TensorFlow.js · Groq API · Llama 3.1 · Chart.js

- Engineered an FMCG simulation platform using **25,000 synthetic personas** to model product launch outcomes before market.
- Trained a neural network for demand prediction and applied **K-Means clustering** to segment customers by behavior, estimating churn rate and customer lifetime value (CLV).

SaviHer — *Women's Safety App (Startup, Beta)* [GitHub] Sep. 2024 – Present

Flutter · Dart · Python · Twilio · Firebase · GCP

- Founded and developed **SaviHer**, **incubated at Bhamasha Techno Hub** a cross-platform Flutter app with an SOS trigger that captures a photo and alerts emergency contacts in **under 2.5 seconds**.
- Added a community volunteering module that cut average help-arrival time by **70%**, and a GCP-based geofencing layer (5 km radius) marking safe vs. unsafe zones.

Achievements

Prize Winner (1,00,000) – AI FOR BHARAT HACKATHON by PAN IIT May 2026

- Recognized as winners for **SaanchaarSetu**, an interoperable middleware layer.

Research Paper Accepted – IDBA 2026 (Springer) Jul. 2026

- *"A Knowledge Graph-Driven Machine Learning Framework for Predictive Market Intelligence and Economic Decision Analytics"* accepted at the 6th International Conference on Data Science and Big Data Analytics (IDBA 2026); to be published in Springer's *Learning and Analytics in Intelligent Systems* series.

Algorand Foundation Hackathon – Finalist & Prize Winner Dec. 2024

- Advanced to Round 3 among international competitors; won **$250**.